

Assignment 2

Course Title: Machine Learning – Session: 2023-2024 (MS)/2020-21 (Undergrad)

Posting date: August 31, 2026

Due Date: September 15, 2026, 11.30 PM

1. [25 pts] Choose a particular type of Data Mining (DM)/Machine Learning (ML) task (e.g., classification and prediction/ clustering) to find its applications. Find at least three articles which present well the application cases matching your selected task and survey theme. An article can be a journal or conference paper (it must provide adequate information about the application problem, the main challenges, and the solution strategy).

Write up a survey report with the following sections:

- i. Topic of the application survey
- ii. Problem description: Define the real-world problem being addressed in the selected papers. Highlight the core domain challenges, data characteristics, and why traditional methods are insufficient.
- iii. Solution technology: Detail the specific DM/ML algorithms, frameworks, and feature engineering techniques used across the three papers. Compare how each paper approached and solved the core technical challenges.
- iv. Solution evaluation: Analyze the performance evaluation methods used in the literature (e.g., metrics like Accuracy, F_1 -score, ROC-AUC, Silhouette Score). Compare the experimental results, strengths, and limitations of the solutions presented in the three papers.
- v. Summary/Conclusion
- vi. References

How to submit:

Prepare a document and write up the survey report in the document. You must add one front page that includes your name, id, course information, and submission information before your answers. Save your final document as Assignment2_yourID.pdf. Now you submit the pdf file to google classroom/CR.